SQL Engineering
HANDBOOK
Learn SQL Like an Engineer

SQL ENGINEERING HANDBOOK

Interview Resources & Roadmaps

Mastering technical screens and system design

🎯 Interview Resources

Part of the Resources Library — SQL Engineering Handbook

This is the file `README.md` points to for interview prep. It's structured as a staged plan, not a flat link list — a SQL interview at each level tests something specific, and the plan below is built around that, not around "here are 40 links, good luck."

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Interview Roadmap

Four stages, each building on the last:

1. **Syntax fluency** — write correct joins, aggregations, and subqueries without hesitation, on a shared screen, while talking.
2. **Pattern recognition** — recognize a "running total," "gaps and islands," or "top-N per group" problem within the first 30 seconds of reading it.
3. **Communication** — narrate your approach *before* writing the query, not after. Interviewers are grading the reasoning as much as the SQL.
4. **Business framing** — connect the query back to the business question it's answering, the way every module in this Handbook is written.

Pair this file with this repo's own `17_SQL_INTERVIEW_QUESTIONS` module and `exercises/interview` directory — those give you Handbook-schema practice; this file gives you the platforms and plan to go wider.

Interviews by Level

Level	What Gets Tested	Prepare With
Beginner	`SELECT`, `WHERE`, basic joins, `GROUP BY`/`HAVING` — correctness and syntax fluency	`01_FUNDAMENTALS`–`03_JOINS`, Coding Platforms below
Intermediate	CTEs, window functions, subqueries, `CASE WHEN` logic, explaining your own query out loud	`04_CASE_WHEN`–`09_DATE_FUNCTIONS`, Practice Websites below
Senior SQL	Query optimization reasoning, index trade-offs, schema critique, defending design decisions	`15_INDEXES`, `16_QUERY_OPTIMIZATION`, `books.md` → Query Optimization
Analytics Engineer	Data modeling (star schema), metric definitions, testing philosophy, dbt-style thinking	`18_SQL_BUSINESS_CASE_STUDIES`, `books.md` → Data Warehousing, `documentation.md` → dbt
Data Engineer	Distributed query behavior, partitioning, pipeline-adjacent SQL, platform-specific dialects	`13_SET_OPERATORS`–`16_QUERY_OPTIMIZATION`, `documentation.md` → Cloud Data Warehouses
FAANG-style	All of the above, under tighter time pressure, often with an unfamiliar schema handed to you cold	Full roadmap, then timed practice on Mock Interview Platforms below

System Design

SQL interviews at the Senior and Data Engineer level increasingly fold in system-design questions — "how would you shard this table," "would you denormalize here." This isn't a separate skill from SQL; it's SQL judgment applied at a larger scale.

- Review `youtube.md` → **ByteByteGo** for how database choices fit into a larger system design answer.

- Review `books.md` → *Designing Data-Intensive Applications* for the reasoning behind partitioning and replication trade-offs you'll be asked to defend.
- Practice narrating trade-offs out loud — "I'd denormalize here because reads outnumber writes 100:1" is a stronger answer than a diagram alone.

Behavioral Preparation

SQL interviews aren't purely technical — expect at least one "tell me about a time" question, usually about a data quality problem you found or a decision you had to defend.

- Prepare 2–3 stories using the STAR structure (Situation, Task, Action, Result), pulled from real project work — your own `projects/` work in this repo is legitimate material to draw from.
- Practice the story that ends in "I was wrong, and here's what I did about it" — interviewers specifically listen for this one, and most candidates don't have it ready.

Platforms

Coding Platforms

Field	Details
Title	LeetCode — Database category
Category	Coding Platforms
Difficulty	Beginner → Advanced (tiered)
Best For	Timed, graded SQL problems in the exact format many companies use for take-home or live-coding rounds
Why It Is Recommended	The most widely referenced platform for SQL interview practice — many companies' actual interview questions are directly inspired by problems here
Free or Paid	Free tier available; premium unlocks company-tagged questions
Who Should Read It	Anyone in active interview prep
Related SQL Handbook Modules	`17_SQL_INTERVIEW_QUESTIONS`

Field	Details
Title	HackerRank — SQL domain
Category	Coding Platforms

Field	Details
Difficulty	Beginner → Advanced
Best For	Structured, certificate-backed SQL practice tracks
Why It Is Recommended	Clear difficulty progression and official skill certificates that are recognized on LinkedIn and by some recruiters
Free or Paid	Free
Who Should Read It	Beginners wanting a structured track with a credential at the end
Related SQL Handbook Modules	`01_FUNDAMENTALS–09_DATE_FUNCTIONS`

Practice Websites

Field	Details
Title	StrataScratch
Category	Practice Websites
Difficulty	Intermediate → Advanced
Best For	Real interview questions sourced from specific companies, with a data-science/analytics lean
Why It Is Recommended	Questions are tagged by the actual company that reportedly asked them, which is useful for targeted prep ahead of a specific interview
Free or Paid	Free tier available; premium unlocks full company-tagged question bank
Who Should Read It	Analytics/data science interview candidates
Related SQL Handbook Modules	`18_SQL_BUSINESS_CASE_STUDIES`

Field	Details
Title	DataLemur
Category	Practice Websites
Difficulty	Beginner → Advanced
Best For	SQL and product-analytics interview questions with a clean, distraction-free practice interface

Field	Details
Why It Is Recommended	Strong free tier with company-tagged questions and clear worked solutions, good for daily short practice sessions
Free or Paid	Free tier available; premium unlocks more questions
Who Should Read It	Anyone doing short, consistent daily practice sessions
Related SQL Handbook Modules	`17_SQL_INTERVIEW_QUESTIONS`

Mock Interview Platforms

Field	Details
Title	Pramp
Category	Mock Interview Platforms
Difficulty	Intermediate → Advanced
Best For	Free, live, peer-to-peer mock interviews with real-time feedback
Why It Is Recommended	Practicing out loud, under time pressure, in front of another person is a different skill than solving a problem alone — this is the free way to get that specific rep in
Free or Paid	Free
Who Should Read It	Candidates within 2–4 weeks of a real interview
Related SQL Handbook Modules	`17_SQL_INTERVIEW_QUESTIONS`

Interview Question Collections

Field	Details
Title	Interview Query
Category	Interview Question Collections
Difficulty	Intermediate → Advanced
Best For	Curated question banks combined with data-role-specific career content (not just SQL in isolation)

Field	Details
Why It Is Recommended	Covers the full interview loop context (SQL, product sense, statistics) rather than SQL as an isolated skill, useful once you're past pure syntax practice
Free or Paid	Free tier available; premium unlocks full content
Who Should Read It	Analytics/data science candidates preparing for a full interview loop, not just the SQL round
Related SQL Handbook Modules	`18_SQL_BUSINESS_CASE_STUDIES`

SQL Cheat Sheets

The primary cheat sheet is this repo's own `20_SQL_CHEATSHEET` and the topic-specific quick references in `cheatsheets/` (joins, CTEs, windows, dates, strings, aggregation) — built against the same schema you've already been practicing on, so nothing needs translating.

> [!TIP] Print or screenshot the cheat sheet the night before an interview. Don't try to memorize syntax you can otherwise look up in five seconds during actual work — memorize the *patterns* (see Interviews by Level) instead.

Business Case Studies

Interview questions increasingly wrap SQL inside a business scenario ("find customers at risk of churning") rather than asking for raw syntax. Practice with:

- This repo's `18_SQL_BUSINESS_CASE_STUDIES` module
- The end-to-end portfolio projects in `projects/` — particularly `hr-analytics` and `ecommerce`, which map closely to common interview scenario types
- `books.md` → *SQL Cookbook*, for pattern recognition across scenario types

Common Mistakes

The same handful of mistakes account for most lost points at every level:

- **Confusing `WHERE` and `HAVING`** — filtering before aggregation vs. after it. Say out loud which one you're using and why.
- **Forgetting a `GROUP BY` column** — every non-aggregated column in the `SELECT` list needs to be in the `GROUP BY`, or the query is wrong even if it happens to run.
- **NULL comparison errors** — `= NULL` silently returns nothing; `IS NULL` is required. This trips up even experienced candidates under pressure.

- **Not narrating the approach first** — jumping straight to typing without stating the plan reads as guessing, even when the final query is correct.
- **Ignoring edge cases out loud** — a strong answer mentions duplicate rows, `NULL`s, or empty result sets even if the final query doesn't fully handle them; it signals awareness.
- **Overcomplicating with nested subqueries** when a CTE would be clearer — interviewers are also grading readability, not just correctness.

Practice Plans

Recommended Weekly Practice Plan

Day	Focus
Mon–Wed	1–2 problems/day from Coding Platforms, one level above comfortable
Thu	Review this repo's `18_SQL_BUSINESS_CASE_STUDIES` — one full case study
Fri	1 mock interview (Pramp) or explain a solved problem out loud, recorded
Weekend	Rest, or light review of `20_SQL_CHEATSHEET` — no new material

30-Day Interview Plan

For: candidates with solid fundamentals who need to sharpen interview-specific execution.

- **Week 1:** Beginner–Intermediate problems daily; drill `Common Mistakes` above until they stop happening.
- **Week 2:** Intermediate–Advanced problems; start narrating solutions out loud before typing.
- **Week 3:** Company-tagged questions on StrataScratch/DataLemur if a specific employer is known; 2 mock interviews.
- **Week 4:** Timed full-length practice sessions; behavioral story prep; final cheat-sheet review.

60-Day Interview Plan

For: candidates who need to build from Intermediate to Senior-level SQL judgment, not just execute faster.

- **Weeks 1–2:** Complete Intermediate Roadmap in [`README.md`](README.md) if not already solid.
- **Weeks 3–4:** `16_QUERY_OPTIMIZATION` + `books.md` → Query Optimization category; start explaining index trade-offs out loud.
- **Weeks 5–6:** Follow the 30-Day Interview Plan above, compressed slightly, now with optimization reasoning folded in.
- **Weeks 7–8 (overlap intentional):** Heavy mock-interview cadence — 2–3/week — plus behavioral prep.

90-Day Interview Plan

For: candidates starting from the Beginner Roadmap and targeting a Senior or Analytics Engineer role.

- **Weeks 1–4:** Beginner Roadmap in full — see `README.md`.
- **Weeks 5–8:** Intermediate Roadmap in full.
- **Weeks 9–11:** Advanced Roadmap, focused on `15_INDEXES` and `16_QUERY_OPTIMIZATION`.
- **Weeks 12–13:** Compressed 30-Day Interview Plan above, with 3+ mock interviews and full behavioral prep.

This closes out the core six files. Second wave next: `newsletters.md`, `courses.md`, `datasets.md`, `playgrounds.md`, `certifications.md`, `communities.md`, `awesome-tools.md`*. Back to Resources Library*.